Quantum Algorithms

1. Deutsch-Jozza algorithm

we need to feed the whole database to the phone book first.

Oracles (the phone book example ✗, Sudoku ✓)

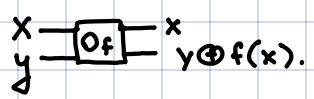
- assume we have access to an oracle, e.g a physical device that we cannot look inside, to which we can pass queries and which return answers.

→ goal: determine some property of the oracle using the minimal number of queries.

on a classical computer, such an oracle is given by a function

$$f: \{0,1\}^n \to \{0,1\}^m$$

on a quantum computer, the oracle must be reversible:

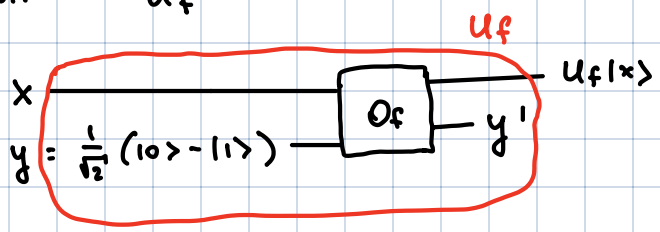


O_f: bit oracle, can be seen as a unitary which perform the map.

→ for $f: \{0,1\}^n \to \{0,1\}$, we can contruct U_f

mod 2 addition

$$O_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$



$$O_f|x\rangle|y\rangle = \frac{1}{\sqrt{2}}\left(O_f|x\rangle|0\rangle - O_f|x\rangle|1\rangle\right) = \frac{1}{\sqrt{2}}\left(|x\rangle|0\oplus f(x)\rangle - |x\rangle|1\oplus f(x)\rangle\right) \circledast$$

$$|x\rangle \cdot (-|y\rangle) = -|x\rangle|y\rangle = (-|x\rangle)\cdot|y\rangle$$

$$\circledast \begin{cases} \frac{1}{\sqrt{2}}|x\rangle(|0\rangle - |1\rangle) = |x\rangle|y\rangle, \text{ if } f(x)=0 \\ \frac{1}{\sqrt{2}}|x\rangle(|0\oplus 1\rangle - |\underbrace{1\oplus 1}_{2 \to \text{becomes } 0}\rangle) = \frac{1}{\sqrt{2}}|x\rangle(|1\rangle - |0\rangle) = -|x\rangle|y\rangle, \text{ if } f(x)=1. \end{cases}$$

$|y\rangle$

$-|y\rangle$

so. $O_f|x\rangle|y\rangle = (-1)^{f(x)}|x\rangle|y\rangle$ ⇒ indep. of $|y\rangle$ ⇒ U_f: phase oracle, which perform the map.

$$U_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

Hadamard on n qubits

recall $H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

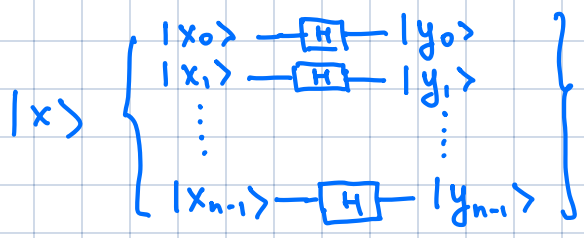
$\Rightarrow$ for $x \in \{0,1\}$:

$$|x\rangle - \boxed{H} - |y\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + (-1)^x |1\rangle\right) \circledcirc$$

$$\circledcirc \frac{1}{\sqrt{2}}\left((-1)^{0\cdot x}|0\rangle + (-1)^{1\cdot x}|1\rangle\right) \circledcirc$$

$$\circledcirc \frac{1}{\sqrt{2}} \sum_{k\in\{0,1\}} (-1)^{k\cdot x} |k\rangle$$

$\Rightarrow$ the above can be Generalized for n qubits.



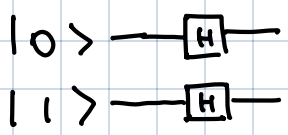
$$|y\rangle = H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k\in\{0,1\}^n} (-1)^{k\cdot x} |k\rangle$$

inner product.

$\rightarrow$ every $|y_i\rangle$ is either $|+\rangle$ or $|-\rangle$

$\hookrightarrow$ $|y\rangle$ must be in a superposition of all possible 2^n bit strings.

e.g $|x\rangle = |01\rangle$



$$|y\rangle = \frac{1}{\sqrt{4}}\Big[(-1)^{(00)\cdot\binom{0}{1}}|00\rangle + (-1)^{(01)\cdot\binom{0}{1}}|01\rangle + (-1)^{(10)\binom{0}{1}}|10\rangle \circledcirc$$

$$\circledcirc (-1)^{(11)\cdot\binom{0}{1}}|11\rangle\Big] = \frac{1}{2}\left[|00\rangle - |01\rangle + |10\rangle - |11\rangle\right].$$

Deutsch-Jozsa algorithm

- We are given a function $f: \{0,1\}^n \rightarrow \{0,1\}$ realized by an oracle, of which we know that it is either constant or balanced

Goal: Determine whether f is constant or balanced.

$\boxed{\text{classic solution}}$ we need to ask the oracle at least twice, but if we get twice the output, we need to ask again, ...

$\Rightarrow$ at most $\frac{N}{2} + 1 = 2^{n-1} + 1$ queries

n: # input bits

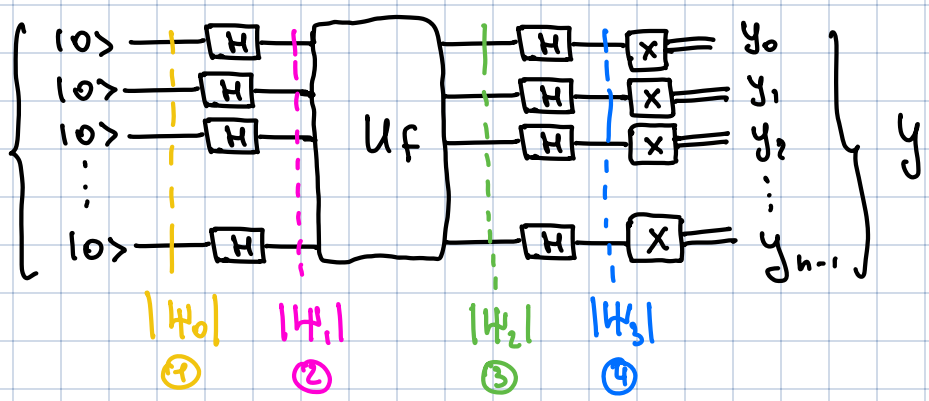
$N = 2^n$: # realisable bit strings.

balanced

half of inputs are mapped to 0, and the other half to 1.

$\boxed{\text{Quantum solution}}$ needs only one query!

circuit for DJ:



Claim: if the outcome y equals the bit string $\underbrace{00\ldots0}$. then f is constant, otherwise it is balanced."

Proof: Let us check the state after each step:

1. $|\psi_0\rangle = |00\ldots0\rangle = |0\rangle^{\otimes n}$

2. $|\psi_1\rangle = H^{\otimes n}|\psi_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} \underbrace{(-1)^{x\cdot \overset{(0\ldots0)}{\psi_0}}}_{+1} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle$

3. $|\psi_2\rangle = U_f \cdot |\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} U_f|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle.$

4. $|\psi_3\rangle = H^{\otimes n}|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum (-1)^{f(x)} \cdot H^{\otimes n}|x\rangle =$

$$= \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} \cdot H^{\otimes n}|x\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} \sum_{k\in\{0,1\}^n} (-1)^{k\cdot x}|k\rangle =$$

$$= \sum_{k\in\{0,1\}^n} \underbrace{\left[\frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)+k\cdot x}\right]}_{\text{Say it is } c_k} |k\rangle =$$

$$= \sum_{k\in\{0,1\}^n} c_k \cdot |k\rangle$$

We do the measurement.

$\Rightarrow$ probability to measure the zero-string is

using Borne rule:

$$\mathbb{P}[y=00\ldots0] = |\langle 00\ldots0|\psi_3\rangle|^2 =$$

$$\circledeq \left| \sum_{k \in \{0,1\}^n} c_k \cdot \langle 00\ldots 01 | k \rangle \right|^2 = |c_{00\ldots 0}|^2 \circledeq$$

$$= \begin{cases} 1, & k = 00\ldots000 \\ 0, & \text{else (orthogonal)} \end{cases}$$

$$\circledeq \left| \frac{1}{2^n} \cdot \underbrace{\sum_{x \in \{0,1\}^n} (-1)^{f(x)}}_{} \right|^2 = \begin{cases} 1 & \text{, if } f(x) \text{ is contant i.e either 0 or 1.} \\ 0 & \text{, if } f(x) \text{ is balanced.} \end{cases}$$

there are 2^n elements here so thats the source of it

$$= \begin{cases} +2^n & \text{if } f(x) \equiv 0 \\ -2^n & \text{if } f(x) \equiv 1 \\ 0 & \text{if } f(x) \text{ balanced} \end{cases}$$